

XUANYU ZHOU

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College of Civil Engineering and Architecture, Zhejiang University

Education

Zhejiang University 09/2021–06/2026 (Expected)

Bachelor of Engineering in Urban and Rural Planning (5-year program)

University of California, Berkeley 01/2025–05/2025

Exchange Program in College of Environmental Design

RESEARCH EXPERIENCE

Research Assistant | MIT-UF-NEU 2025 Joint Summer Research Camp

Supervisor: Xuan Jiang, Ph.D. | 06/2025–10/2025

I participate in the development of a novel, **GPU-based computational framework designed to accurately model large-scale Urban Air Mobility (UAM) demand and optimize the operations of heterogeneous aircraft fleets**. Throughout the project, I played an integral role in the entire research lifecycle, conducting in-depth result analysis, creating data visualizations, and serving as the primary author of the research manuscript, where I synthesized our findings and formulated the final conclusions.

Research Assistant | Monitoring-Mapping-Modeling (3M) lab, UC Berkeley

Supervisor: Prof. Lu Liang | 03/2025–present

Developing urban Sky View Factor (SVF) datasets across the United States using airborne LiDAR data, with validation through comparative analysis of street view imagery. Responsible for processing street view data and conducting preliminary analysis. Collaborating with Ph.D. student Yuye Zhou; project ongoing.

Research Assistant | Qizhenwenxue Research Program, ZJU

Supervisor: Prof. Shuang Ma | 04/2023–Present

Authored the Introduction and Literature Review for **a case study on post-pandemic urban revival in Nagoya**, which leveraged mobile phone signaling data and a Geographic Random Forest (GRF) model. My work established the theoretical framework by synthesizing existing literature on urban mobility and resilience. This research has been published.

Applied a GRF model to analyze a dataset of over 600,000 geotagged Weibo posts in Beijing across four months in an investigation into **the built environment's impact on mental health**. My primary responsibility was conducting the experiments and interpreting the resulting spatial patterns. The manuscript is currently under review.

Lead the data analysis for **exploring the synergy between shared bicycles and the built environment**, processing a robust dataset of over 9 million trip records in Shenzhen using GRF. My role extended beyond generating results to include a thorough analysis of the findings and contributing to the conclusion section. A co-authored manuscript from this work is presently under review.

Group Lead | Rural Revitalization Social Investigation, ZJU

Supervisor: Prof. Yuehua Zhang | 09/2023–11/2023

Spearheaded a 10-member team **conducting field surveys on rural revitalization and common prosperity** in Jiashan, Zhejiang; synthesized key findings into a group report.

RESEARCH & ACADEMIC CONTRIBUTIONS

Peer-Review Articles

Ma, S., **Zhou, X.**, Cai, W., et al. (2023). Evaluating the social-economic recovery impacts of built environment for post-pandemic. *Transaction in urban data, science, and technology*. DOI: <https://doi.org/10.1177/27541231251314132>

Ma, S., Li, W., **Zhou, X.**, Ma, T., He, Y., and Li, S. (Manuscript under review). Future life expectancy extension under bike ride, air pollution exposure, and traffic accidents through modifying the built environment and climate: An integrated modelling study. *npj Urban Sustainability*.

Li, S., Cai, W., Wu, C., Zhang, X., **Zhou, X.**, He, Y., and Ma, S. (Manuscript under review). Exploring the relationship between resident emotions and built environment in Beijing based on a geographical weighted random forest approach. *Sustainable Cities and Society*.

Zhou, X., & Chen, M. (Manuscript under review). Greening the Gap: Examining Urban Greenery Equity in Shrinking Cities. *Urban Forestry & Urban Greening*.

Jiang, X., **Zhou, X.**, Zhao, Y., et al. (Manuscript under review). From Patchwork to Network: A Comprehensive Framework for Demand Analysis and Fleet Optimization of Advanced Air Mobility. *IEEE Transactions in Intelligent Transportation System*.

Conference Presentations

Zhou, X., & Chen, M. (2024). Greening the Gap: Examining Urban Greenery Equity in Shrinking Cities. 2024 ACSP Annual Conference (Oral Presentation).

Jiang, X., **Zhou, X.**, Zhao, Y., et al. (2026). FROM PATCHWORK TO NETWORK: A Comprehensive Framework for Demand Analysis and Fleet Optimization of Urban Air Mobility. 2026 Transportation Research Board Annual Meeting (Accepted). DOI: <https://doi.org/10.48550/arXiv.2510.04186>

Competitions

2024 WUPENiCity Competition: Led research on Park and Ride (P+R) Development Strategies for Urban Railway Stations, overseeing data collection, processing, mapping, and drafting the final report.

HONORS & AWARDS

Zhejiang Provincial Government Scholarship, Zhejiang Province (2024, 2025)

Second-Prize Scholarship, Zhejiang University (2023, 2024, 2025)

Third-Prize Scholarship, Zhejiang University (2022)

Jincheng Academic Scholarship, Zhejiang University (2023)

Academic Excellence Award, Zhejiang University (2022, 2023, 2024, 2025)

Outstanding Student Leader, Zhejiang University (2022, 2023, 2024)

Outstanding Community Service Volunteer, Zhejiang University (2025)

Distinguished Investigator, Innovation Center of Yangtze River Delta (2023)

TECHNICAL SKILLS

Programming Languages: Skilled in Python and R for data analysis and modeling.

Geographic Information Systems (GIS): Experienced in spatial data analysis and visualization using GIS tools.

3D Modeling Software: Skilled in Rhino and SketchUp for modeling.

Deep Learning: Entry-level experience in deep learning and AI applications.

Research Tools: Proficient in literature review, and academic writing.